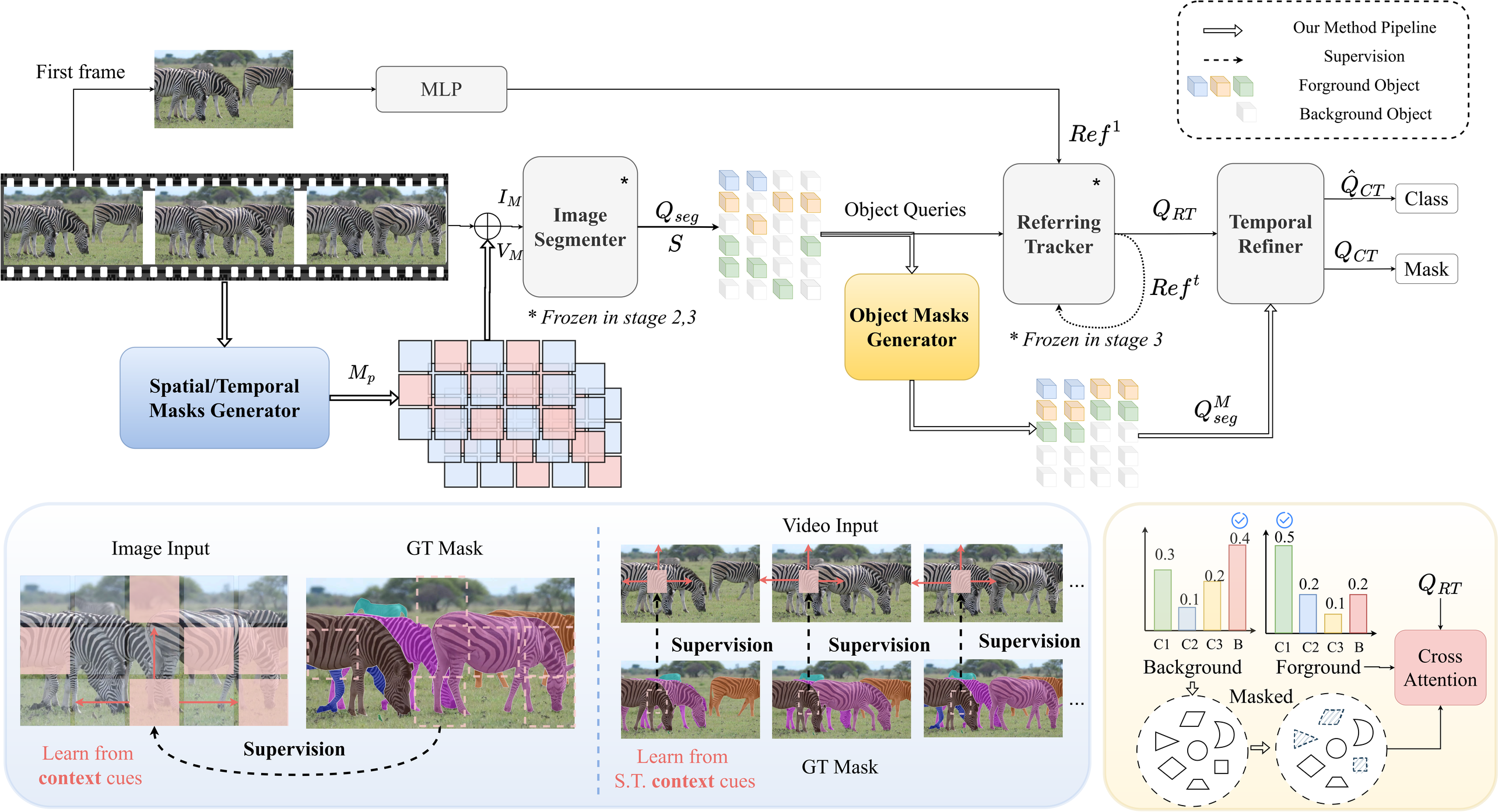
Our Method Pipeline
Supervision
Forground Object
Background Object
First frame
MLP
Ref^1
I_M
V_M
Image Segmenter
*
Q_{seg}
S
Object Queries
Referring Tracker
*
Q_{RT}
Temporal Refiner
$\hat{Q}_{CT}$
Class
Q_{CT}
Mask
Ref^t
* Frozen in stage 2,3
Object Masks Generator
* Frozen in stage 3
Spatial/Temporal Masks Generator
M_p
Q^M_{seg}
Image Input
GT Mask
Supervision
Learn from context cues
Video Input
Supervision
Supervision
Supervision
Learn from S.T. context cues
GT Mask
0.3
0.1
0.2
0.4
C1 C2 C3 B
Background
0.5
0.2
0.1
0.2
C1 C2 C3 B
Forground
Q_{RT}
Cross Attention
Masked